CRAMER'S RULE

LINEAR ALGEBRA

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

Cramer's Rule

CRAMER'S RULE

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_3 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_3 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$A_x = \begin{bmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_3 \\ d_3 & b_3 & c_3 \end{bmatrix}$$

$$A_y = \begin{bmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_3 \\ a_3 & d_3 & c_3 \end{bmatrix}$$

$$A_z = \begin{bmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_3 \\ a_3 & b_3 & d_3 \end{bmatrix}$$

$$A_x = \begin{bmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_3 \\ d_3 & b_3 & c_3 \end{bmatrix}$$

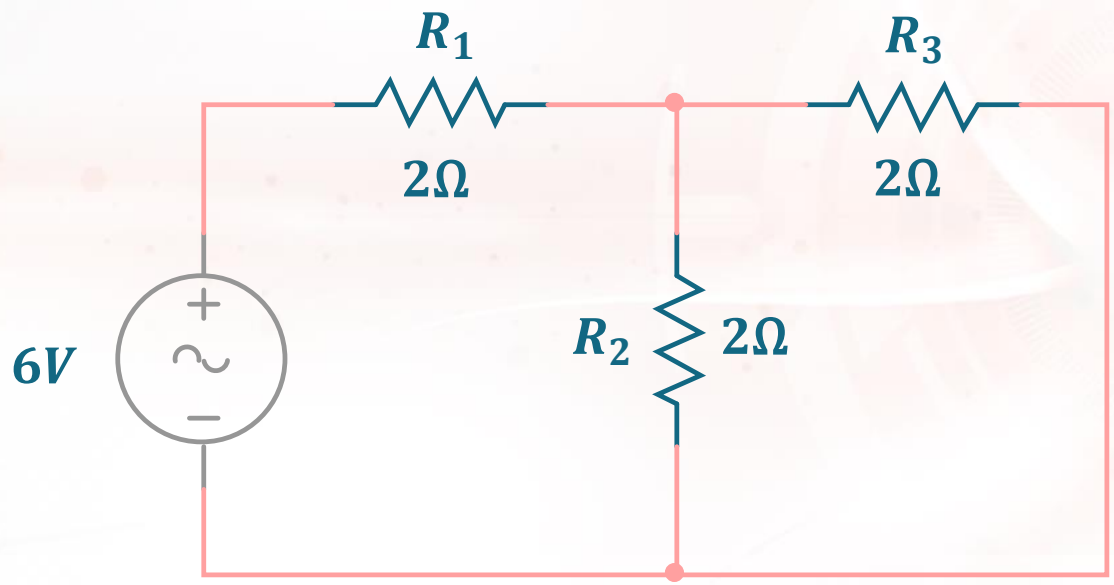
$$x = \frac{|A_x|}{|A|} \qquad y = \frac{|A_y|}{|A|} \qquad z = \frac{|A_z|}{|A|}$$

$$A_y = \begin{bmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_3 \\ a_3 & d_3 & c_3 \end{bmatrix}$$

$$A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_3 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$A_z = \begin{bmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_3 \\ a_3 & b_3 & d_3 \end{bmatrix}$$

Determine the voltages v_1, v_2 and v_3 of the given network.



$$-\frac{v_1}{2} + \frac{v_2}{2} + \frac{v_3}{2} = 0$$

$$v_1 + v_2 + 0 = 6$$

$$0 - v_2 + v_3 = 0$$

SEATWORK